

Switch FI: Evaluating Memory-Augmented Grounded Reasoning for Personalized Financial LLM Assistance

Author Name*, Author Name[†], Author Name[‡]

*Department / Institution
City, Country
email@example.com

Author Name[§]

§Department / Institution
City, Country
email@example.com

Abstract—Large language models (LLMs) can provide useful financial guidance but remain unreliable when answering questions requiring access to a user’s actual financial records or persistent knowledge of user-specific preferences. In particular, ungrounded LLM responses can hallucinate numerical values, while stateless assistants repeatedly lose information about user preferences and financial goals.

We present *Switch FI*, a Model Context Protocol (MCP)-based financial assistance system that combines deterministic financial aggregation with explicit and implicit user memory. The system connects LLM assistants to consented financial data through India’s Account Aggregator (AA) ecosystem and exposes structured financial operations as callable tools. Rather than allowing the language model to directly infer numerical financial state, Switch FI performs deterministic aggregation over transaction data and supplies the resulting values to the model as grounded context.

We further introduce an evaluation framework for measuring the effect of persistent memory on personalized financial reasoning. The evaluation compares memory-enabled and stateless configurations over a fixed set of financial questions spanning numerical grounding, preference adherence, and multi-turn memory retrieval. The experimental dataset contains synthetic but realistic financial records with known ground truth, enabling reproducible evaluation.

Results will be populated after completion of the experimental evaluation.

Index Terms—large language models, financial AI, personalization, memory, grounded reasoning, Model Context Protocol, account aggregator, financial question answering

I. INTRODUCTION

Large language models (LLMs) have demonstrated strong capabilities in natural language understanding, reasoning, and interactive assistance. However, their usefulness in personal finance is constrained by two fundamental limitations: lack of direct access to current financial records and limited persistence of user-specific information.

A user may ask questions such as “How much did I spend on food last month?”, “Am I spending more than usual on subscriptions?”, or “Given my preference to keep an emergency fund of six months, how much should I save this month?”. These questions require more than generic financial knowledge. They require access to user-specific transaction data, deterministic numerical computation, and knowledge

of preferences that may have been expressed during earlier interactions.

Without external grounding, an LLM may produce plausible but incorrect numerical values. Similarly, without persistent memory, an assistant may know that a user previously expressed a financial preference but fail to apply that preference in a later interaction.

Recent work on LLM personalization has demonstrated the importance of memory extraction, retrieval, and utilization. AlpsBench, for example, evaluates personalization through tasks covering information extraction, information update, information retrieval, and information utilization [?]. Its evaluation framework demonstrates that remembering information and successfully applying it are distinct capabilities.

Financial assistance introduces an additional requirement: responses must be grounded in an external source of financial truth. This creates an intersection between two problems:

- 1) **Grounded financial reasoning:** obtaining accurate financial facts from structured financial data.
- 2) **Personalized reasoning:** retrieving and applying user-specific preferences and constraints across interactions.

Switch FI addresses these requirements through a hybrid architecture. Financial state is computed using deterministic tools, while personalization information is represented using structured explicit and implicit memory. The resulting context is then supplied to an LLM through an MCP interface.

The central research question investigated in this work is:

Does persistent user memory improve personalized financial reasoning when the underlying financial facts are deterministically grounded?

This question is investigated using a controlled ablation between memory-enabled and stateless configurations.

A. Contributions

This work makes the following contributions:

- We present the architecture of *Switch FI*, an LLM-agnostic MCP server for grounded personal financial assistance.
- We introduce a deterministic financial reasoning layer that computes financial summaries from structured transaction

records rather than relying on LLM-generated numerical calculations.

- We implement a structured memory layer supporting both explicit user preferences and implicitly inferred preferences.
- We define an evaluation protocol separating numerical grounding, preference adherence, and multi-turn memory retrieval.
- We formulate a controlled memory ablation comparing identical financial data and questions under memory-enabled and stateless configurations.

II. RELATED WORK

A. LLM Personalization and Memory

Personalization in LLM systems has increasingly shifted from static prompt-based personalization toward persistent memory mechanisms. Recent benchmarks distinguish between systems that simply retrieve historical context and systems that explicitly evaluate the lifecycle of personalized information.

AlpsBench evaluates four stages of personalization: personalized information extraction, personalized information update, personalized information retrieval, and personalized information utilization [?]. The benchmark further distinguishes multiple forms of personalization, including preference following and constraint following.

This distinction is important for financial assistants. A system may correctly remember that a user prefers conservative spending but still fail to apply that preference when making a recommendation.

Existing personalization research therefore motivates evaluating memory not merely as storage, but as a mechanism that influences downstream reasoning.

B. Financial LLM Grounding

LLM applications in finance introduce additional reliability concerns. Financial questions frequently contain numerical quantities, temporal constraints, and domain-specific terminology. Hallucinated values can therefore result in materially incorrect conclusions.

A common approach to hallucination mitigation is retrieval-augmented generation, where external information is retrieved before generation. However, financial assistance requires more than document retrieval. Transaction-level information must often be aggregated, categorized, and computed before it can be meaningfully provided to an LLM.

Switch FI therefore separates deterministic financial computation from language generation.

C. Consent-Based Financial Data Access

India’s Account Aggregator ecosystem provides a consent-driven architecture for accessing financial information from participating financial institutions. This model enables financial information to be shared with a requesting entity based on user authorization.

Switch FI uses the Account Aggregator model as the data-access boundary. For experimentation, the system uses a Finvu sandbox adapter and a synthetic/mock data mode.

This architecture separates:

- 1) financial data acquisition,
- 2) financial computation,
- 3) personalization memory, and
- 4) natural-language reasoning.

D. Research Gap

Existing personalization benchmarks primarily study general-purpose conversational interactions, while financial LLM systems often focus on grounding and hallucination reduction.

Switch FI combines these dimensions by evaluating whether memory affects personalized reasoning *after* financial facts have been grounded through deterministic computation.

III. SWITCH FI SYSTEM

A. System Overview

Switch FI is designed as an intermediary between an LLM assistant and a user’s financial information.

The architecture consists of four major layers:

- 1) **Consent and data acquisition layer**
- 2) **Deterministic financial reasoning layer**
- 3) **Memory layer**
- 4) **MCP interface layer**

Figure 1 presents the overall architecture.

B. Consent and Data Acquisition

The financial data acquisition layer abstracts the underlying data provider through an adapter interface.

For production deployment, the adapter is intended to communicate with the Account Aggregator ecosystem. During development and experimentation, a mock adapter generates deterministic synthetic transaction data.

The system supports two modes:

$$D = \begin{cases} D_{\text{mock}}, & \text{if MOCK_MODE} = \text{true} \\ D_{\text{AA}}, & \text{otherwise} \end{cases} \quad (1)$$

where D_{mock} represents synthetic financial data and D_{AA} represents data obtained through the Account Aggregator integration.

The mock dataset is generated with known ground truth so that numerical answers can be evaluated without ambiguity.

C. Deterministic Financial Reasoning

A central design principle of Switch FI is that the LLM should not be responsible for computing authoritative financial totals.

Let the user’s transaction dataset be

$$T = \{t_1, t_2, \dots, t_n\}. \quad (2)$$

Each transaction t_i contains fields including amount, timestamp, merchant, and category.

figures/system_architecture.pdf

Fig. 1. Switch FI system architecture. Financial data is acquired through a consent-based adapter, transformed into deterministic financial summaries, combined with relevant user memory, and exposed to an LLM through MCP tools.

A deterministic aggregation function computes a financial state:

$$z = f(T), \quad (3)$$

where z may contain:

- total income,
- total expenditure,
- category-wise expenditure,
- recurring expenses,
- monthly summaries,

- savings,
- account balances, and
- other derived financial quantities.

The resulting structured output is then supplied to the LLM. This creates the following separation:

Financial Truth $\rightarrow$ Deterministic Computation $\rightarrow$ LLM Explanation. (4)

For transaction categorization, the system uses a regex/rule-based strategy for deterministic matches, followed by an LLM

fallback for ambiguous transactions.

D. Memory Architecture

Switch FI maintains structured user memory independently from raw financial transactions.

For a user u , memory is represented as:

$$M_u = \{m_1, m_2, \dots, m_k\}. \quad (5)$$

Each memory record may contain:

- memory content,
- memory type,
- source,
- confidence,
- creation timestamp,
- update timestamp.

We distinguish two memory categories.

1) *Explicit Memory*: Explicit memory represents information directly stated by the user.

Examples include:

- “I want to maintain a six-month emergency fund.”
- “I prefer not to spend more than INR 5,000 on subscriptions.”
- “I am saving for a car.”

2) *Implicit Memory*: Implicit memory represents preferences or behavioral information inferred from repeated interactions.

Examples include recurring spending preferences or consistently expressed financial constraints.

The distinction between explicit and implicit memory is motivated by prior work showing that real-world user interactions contain both direct and indirect personalization signals [?].

E. Memory Retrieval

Given a user query q , the memory subsystem retrieves a relevant subset:

$$R_q = R(M_u, q). \quad (6)$$

The final LLM context is therefore represented as:

$$C = \{q, z, R_q\}, \quad (7)$$

where:

- q is the user query,
- z is deterministic financial state, and
- R_q is retrieved personalized memory.

The LLM generates the final response:

$$y = LLM(C). \quad (8)$$

TABLE I
EVALUATION DATASET CONFIGURATION

Property	Value
Number of users	[TBD]
Number of accounts	[TBD]
Number of months	[TBD]
Number of transactions	[TBD]
Number of categories	[TBD]
Evaluation questions	[TBD]
Known ground truth	Yes

F. Model Context Protocol Interface

Switch FI exposes financial and memory operations through the Model Context Protocol (MCP).

The MCP interface provides tools for operations such as:

- retrieving financial summaries,
- retrieving transactions,
- calculating category expenditure,
- querying monthly spending,
- retrieving user memories,
- storing explicit preferences,
- updating memory, and
- obtaining financial context for reasoning.

This makes the financial infrastructure independent of the specific LLM client.

IV. EVALUATION FRAMEWORK

A. Research Questions

The evaluation is organized around four research questions.

RQ1: Does persistent memory improve preference adherence in personalized financial responses?

RQ2: Does memory affect numerical grounding accuracy when financial facts are deterministically computed?

RQ3: Does persistent memory improve retrieval of information expressed earlier in a conversation?

RQ4: Does memory change the confidence or calibration behavior of the LLM when answering financial questions?

B. Dataset Construction

The evaluation uses a synthetic but realistic financial dataset generated through the Switch FI mock adapter.

Each synthetic user is associated with one or more financial accounts containing transactions across multiple months.

The dataset is intentionally constructed so that every numerical quantity required by an evaluation question has a known ground-truth value.

C. Evaluation Configurations

The primary experiment is an ablation comparing two configurations.

Both configurations use:

- identical financial datasets,
- identical question sets,
- identical deterministic financial tools,
- identical LLM,

TABLE II
EXPERIMENTAL CONFIGURATIONS

Configuration	Memory	Financial Tools
Memory-Off	Disabled	Enabled
Memory-On	Enabled	Enabled

- identical system instructions.

The only manipulated variable is persistent memory availability.

The evaluation model is:

$$\text{Claude Haiku 4.5 via OpenRouter} \quad (9)$$

with the exact model identifier to be reported in the final experiment.

D. Evaluation Tasks

The question set contains approximately 30–50 fixed questions divided into three primary task categories.

1) *Task 1: Numerical Grounding*: For a financial query q , the system produces an answer $\hat{a}$. The expected numerical answer is obtained from deterministic ground truth a^* .

The numerical evaluation therefore measures:

$$Acc_{\text{num}} = \frac{\sum_{i=1}^N \mathbb{I}[\hat{a}_i = a_i^*]}{N}. \quad (10)$$

For values where rounding or formatting differences are expected, an absolute tolerance ϵ may be used:

$$\mathbb{I}[|\hat{a}_i - a_i^*| \leq \epsilon]. \quad (11)$$

2) *Task 2: Preference Adherence*: Preference adherence evaluates whether the response respects a known user preference or constraint.

For question q_i with expected preference behavior p_i , let

$$A_i = \begin{cases} 1 & \text{if the response satisfies } p_i, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

Preference adherence rate is:

$$PAR = \frac{1}{N} \sum_{i=1}^N A_i. \quad (13)$$

3) *Task 3: Memory Retrieval*: For multi-turn questions, the assistant must retrieve information introduced in an earlier interaction.

Let m^* denote the ground-truth memory and $\hat{m}$ the information used by the model.

Memory retrieval accuracy is:

$$Acc_{\text{mem}} = \frac{\sum_{i=1}^N \mathbb{I}[\hat{m}_i = m_i^*]}{N}. \quad (14)$$

Semantic matching may additionally be used where exact lexical matching is inappropriate.

E. Confidence Calibration

The assistant may additionally be asked to provide a confidence label for each answer.

Let c_i denote the predicted confidence and r_i indicate whether the answer is correct.

The relationship between confidence and correctness will be analyzed using:

$$Calibration = \frac{1}{N} \sum_{i=1}^N |c_i - r_i|. \quad (15)$$

Lower values indicate closer agreement between stated confidence and observed correctness.

V. EXPERIMENTAL RESULTS

A. Overall Results

The primary comparison between memory-enabled and memory-disabled configurations is shown in Table IV.

Important: all values in this section remain placeholders until the experimental harness has been executed.

B. Numerical Grounding

The numerical grounding experiment measures whether introducing memory changes the model’s ability to answer questions whose ground truth is derived from deterministic financial aggregation.

C. Preference Adherence

The second experiment evaluates whether persistent memory improves the assistant’s ability to follow financial preferences.

D. Memory Retrieval

Memory retrieval evaluates whether information introduced in an earlier turn is correctly used in a subsequent financial question.

E. Ablation Analysis

The principal ablation compares the same model under two memory conditions.

The relative improvement of memory-enabled reasoning is calculated as:

$$\Delta_{\text{rel}} = \frac{Score_{\text{on}} - Score_{\text{off}}}{Score_{\text{off}}} \times 100. \quad (16)$$

VI. DISCUSSION

A. Does Memory Improve Financial Personalization?

The primary purpose of the experiment is to determine whether persistent memory improves personalized financial reasoning.

If the Memory-On configuration produces a higher preference adherence rate while maintaining comparable numerical accuracy, this would suggest that memory primarily improves personalization rather than financial grounding itself.

Observed result: [TBD after evaluation].

TABLE III
EVALUATION TASK CATEGORIES

Task	Description	Example
Numerical Grounding	Tests whether the assistant returns the correct numerical value derived from financial records.	“How much did I spend on food last month?”
Preference Adherence	Tests whether stored financial preferences influence recommendations.	“How much should I allocate to discretionary spending given my savings goal?”
Memory Retrieval	Tests whether information introduced earlier is correctly recalled and used in a later interaction.	“What savings goal did I tell you I was working toward?”

TABLE IV
OVERALL EVALUATION RESULTS

Metric	Memory-Off	Memory-On
Numerical Accuracy	[TBD]	[TBD]
Preference Adherence	[TBD]	[TBD]
Memory Retrieval	[TBD]	[TBD]
Confidence Calibration	[TBD]	[TBD]

TABLE V
NUMERICAL GROUNDING PERFORMANCE

Category	Off	On	Δ
Income	[TBD]	[TBD]	[TBD]
Expenses	[TBD]	[TBD]	[TBD]
Categories	[TBD]	[TBD]	[TBD]
Savings	[TBD]	[TBD]	[TBD]
Overall	[TBD]	[TBD]	[TBD]

TABLE VI
PREFERENCE ADHERENCE RESULTS

Preference Type	Off	On	Δ
Budget Constraint	[TBD]	[TBD]	[TBD]
Savings Goal	[TBD]	[TBD]	[TBD]
Spending Preference	[TBD]	[TBD]	[TBD]
Risk Preference	[TBD]	[TBD]	[TBD]
Overall	[TBD]	[TBD]	[TBD]

TABLE VII
MULTI-TURN MEMORY RETRIEVAL

Task	Off	On	Δ
Explicit Memory	[TBD]	[TBD]	[TBD]
Implicit Memory	[TBD]	[TBD]	[TBD]
Multi-Hop Recall	[TBD]	[TBD]	[TBD]
Overall	[TBD]	[TBD]	[TBD]

B. Memory Versus Financial Grounding

The architecture deliberately separates memory from financial grounding.

Deterministic financial tools provide:

$$\text{Ground Truth}_{\text{financial}} \quad (17)$$

while memory provides:

$$\text{Context}_{\text{personal}}. \quad (18)$$

The LLM combines these two sources during response generation.

This distinction allows the experiment to determine whether memory introduces improvements independently of numerical grounding.

C. Comparison With Existing Personalization Benchmarks

The evaluation design is inspired by the observation in recent personalization benchmarks that remembering information and utilizing that information are separate capabilities [?].

Unlike general conversational personalization benchmarks, Switch FI introduces a deterministic financial state as an additional source of ground truth.

The resulting evaluation can therefore be viewed as:

$$\text{Personalization} + \text{Grounded Financial Reasoning}. \quad (19)$$

This combination is the central distinction of the proposed framework.

D. Potential Failure Modes

Several failure modes are expected to be particularly relevant:

- 1) The model retrieves the correct memory but fails to apply it.
- 2) The model applies a preference that conflicts with newer information.
- 3) The model correctly retrieves financial data but performs incorrect reasoning over the values.
- 4) The model generates unsupported numerical values despite receiving deterministic tool output.
- 5) Implicit memory is inferred incorrectly from limited evidence.

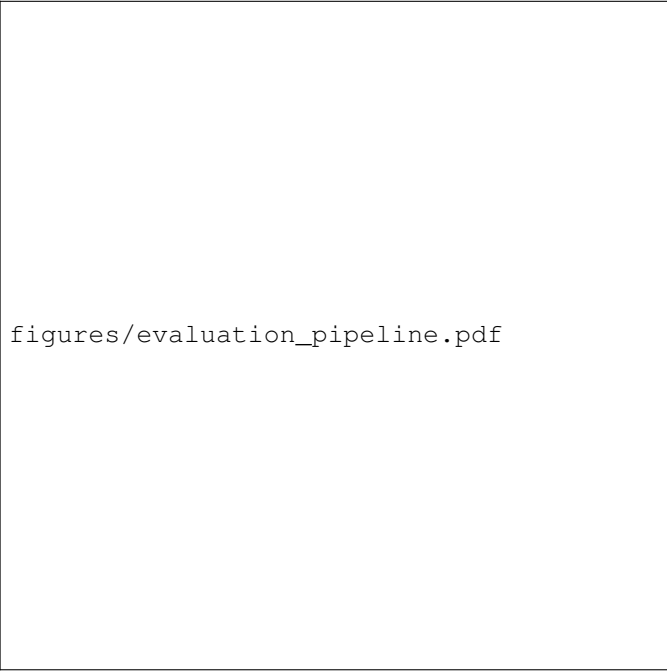
These failure modes should be analyzed separately rather than collapsing all errors into a single accuracy value.

VII. LIMITATIONS

This study has several limitations.

A. Synthetic Evaluation Data

The experimental financial dataset is generated synthetically. Although the dataset is designed to resemble realistic financial transactions, it does not represent the complete variability of real-world banking data.



figures/evaluation_pipeline.pdf

Fig. 2. Evaluation pipeline comparing memory-enabled and memory-disabled Switch FI configurations using identical financial data and question sets.

B. Sandbox Data Access

The current implementation uses the Finvu sandbox and a mock adapter rather than production financial accounts.

Consequently, the evaluation does not establish the operational performance of the system against production-scale banking data.

C. Limited Model Coverage

The initial evaluation uses a single LLM configuration. Results may therefore depend on model-specific reasoning and instruction-following behavior.

Future experiments should evaluate multiple models under identical conditions.

D. Evaluation Scale

The current evaluation uses a relatively small fixed question set. Although this provides controlled paired comparisons, a larger dataset would provide stronger statistical confidence.

E. Automated Evaluation

Some evaluation metrics rely on automated grading or deterministic matching. Human evaluation may be required for ambiguous cases involving preference interpretation and financial recommendations.

VIII. CONCLUSION

This paper presented Switch FI, an MCP-based architecture for memory-augmented and financially grounded LLM assistance.

The system separates financial truth from language generation by performing deterministic aggregation over structured

transaction data, while a persistent memory layer provides user-specific preferences and context.

We introduced an evaluation framework that compares memory-enabled and stateless configurations across numerical grounding, preference adherence, and multi-turn memory retrieval.

The final experimental results will determine whether persistent memory provides measurable improvements in personalized financial reasoning without degrading numerical grounding.

Final empirical finding: [TBD].

Future work includes integration with production Account Aggregator credentials, evaluation across multiple LLMs, larger real-world user studies, improved implicit-memory handling, and deployment of Switch FI as a production financial MCP service.

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